

John C. Glenn II

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EDUCATION

M.S. Mechanical Engineering | NC State University | GPA: 4.0 May 2026
B.S. Mechanical Engineering | NC State University | GPA: 3.65 May 2025
Undergraduate Coursework | UNC Wilmington | GPA: 3.97 August 2020 - May 2022

EXPERIENCE

NC State Plant Sciences Initiative | Mechanical Engineering Intern Raleigh, NC | May 2025 - May 2026

- Led the mechanical design of a modular agricultural robot and formulated its GNC architecture, simulating a Nonlinear Model Predictive Controller (NLMPC) using CasADi for precise trajectory tracking
- Implemented end effector control in Python via an object-oriented hardware abstraction layer for stepper motors

Gamewell Mechanical | Mechanical Engineering Intern Salisbury, NC | May 2024 - August 2024

- Identified an inefficiency in ductwork labeling and designed and implemented a custom tool that reduced process time and labor costs by over 50% while eliminating the need for scissor lifts

Martin Marietta Materials | Mechanical Engineering Intern Castle Hayne, NC | May 2023 - August 2023

- Designed a diverting chute to support plant expansion using SolidWorks which improved efficiency and reduced downtime by redirecting flow of material during maintenance periods

PROJECTS

Double Inverted Pendulum on a Cart: Swing-up and Stabilization

- Developed both passivity-based partial feedback linearization and offline trajectory optimization (trapezoidal direct collocation) for underactuated swing-up control, reducing swing-up time from 19.1s to 4.3s
- Designed and tuned online model predictive controller (MPC) to track trajectory and stabilize the pendulum once within the region of attraction (ROA) at desired equilibrium points

Spacecraft Orbit Determination and State Estimation

- Developed an Iterated Square Root Information Filter (SRIF) and Extended Kalman Filter (EKF) from scratch in MATLAB for spacecraft orbit determination with unknown orbital parameters and dynamics
- Estimated spacecraft states, impulsive maneuvers, J2/J3, drag, lunar third-body effects, and ground station positions given only raw range and range-rate measurements and an *a priori* state estimate

Inverted Pendulum for Controls Curriculum

- Engineered and fabricated an inverted pendulum on a cart for NCSU's controls curriculum, prioritizing robustness to withstand repeated student use and controller failures
- Implemented sliding mode control using Simulink Desktop Real-Time (SDRT) and modeled system dynamics
- Designed CAD model and shop drawings in SolidWorks utilizing GD&T standards and DFM principles

Autonomous Tomato Harvester

- Developed a multi-input multi-output (MIMO) LQG control system using optimal control theory for a 3-dof mechanical arm and iris diaphragm gripper
- Achieved end effector positioning accuracy of $\pm 2\text{mm}$ by utilizing a gain scheduled LQR controller for optimal performance at various operating points and a Kalman filter for optimal state estimation
- Applied grey-box system identification techniques to model system dynamics from experimental data
- Identified ripe tomatoes and position data using computer vision, convolutional neural networks, and triangulation
- Applied design for manufacturing (DFM) principles during CAD design and fabrication of components on mill

Robotic Manipulator Simulation and Control

- Simulated pick-and-place manipulator in Mujoco/Python, modeling contact dynamics for reliable grasping
- Implemented inverse kinematics (IK) to map desired end-effector poses to joint space configurations

SKILLS

GNC: LQR, SMC, MPC, Feedback Linearization, Passivity-based Control, Gain Scheduling, Optimal Control, Trajectory Optimization, System Identification, State Estimation (EKF, SRIF), Inverse Kinematics, PID, Algorithm Development

Software: MATLAB, Simulink, SDRT, Python, Git version control, Linux, Mujoco, SolidWorks, Fusion360, FEMM

Hardware: Oscilloscope, Signal Generator, Microcontrollers, Motor Control, HIL Simulation, Vertical Mill, 3D Printing